

Problem 1

What is the probability that a model generated a sequence of observations?

$$O = \{O_1, O_2, \dots, O_T\} \quad \lambda = \{A, B, \lambda\}$$

$$P(O|\lambda)?$$

Theory

Let's start with all possible state sequences

$$Q = \{q_1, q_2, \dots, q_T\}$$

$$\text{Then } P(O|Q, \lambda) = \prod_t P(O_t | q_t, \lambda) \\ = \prod_t b_{q_t}(O_t)$$

$$\text{And } P(Q|\lambda) = \prod_t P(q_t | \lambda) \\ = \prod_{q_1} a_{q_1, q_2} \dots a_{q_{T-1}, q_T}$$

$$\text{Therefore } P(O, Q|\lambda) = P(O|Q, \lambda) P(Q|\lambda)$$

And $P(O|\lambda) = \sum_{\text{all } Q} P(O, Q|\lambda)$ and calculating is infeasible, with order of $(2T-1)N^T + (N^T - 1)$

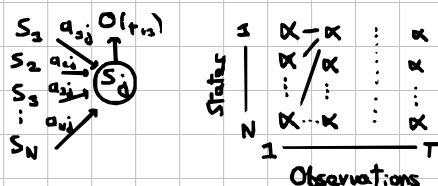
Forward - Backward algorithm

Introducing helper function:

$$\alpha_t(i) = P(O_1, O_2, \dots, O_t, q_t = S_i | \lambda)$$

$$\text{With, } \alpha_1(i) = \pi_i b_i(O_1) \text{ and } \alpha_{t+1}(i) = \sum_j \alpha_t(j) a_{ji} b_i(O_{t+1})$$

$$\text{Therefore } P(O|\lambda) = \sum_i \alpha_T(i)$$



Also works backwards with helper:

$$\beta_t(i) = P(O_{t+1}, O_{t+2}, \dots, O_T | q_t = S_i, \lambda)$$

$$\text{With, } \beta_T(i) = 1 \text{ and } \beta_t(i) = \sum_j a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)$$

$$\text{Therefore } P(O|\lambda) = \sum_i \beta_1(i)$$

